Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. ***Time: 03 Hours***

Code: 104304 ***Subject: Network Theory*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) What is the value of the unit step function u (t – T) at any time t ≤ T
(i) 1 (ii) 0 (iii) infinite (iv) none of the above

(b) An RC high-pass filter consists of a 820 ohm resistor. What is the value of C is to be taken so that X_C is ten times less than R at an input frequency of 12 kHz?
(i) 81 μF (ii) 161 μF (iii) 0.161 μF (iv) 220 μF

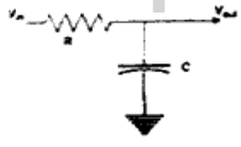
(c) The transient current in an RLC circuit is oscillating when
(i) $R = 2\sqrt{L/C}$ (ii) R=0 (iii) $R < 2\sqrt{L/C}$ (iv) $R > 2\sqrt{L/C}$

(d) A cut set matrix gives relation between
(i) Branch currents and link currents
(ii) Branch voltages and tree branch voltages
(iii) Branch voltages and link voltages
(iv) Branch currents and tree currents

(e) A network consists 4 nodes, 6 branches. How many basic loop can be formed?
(i) 3 (ii) 9 (iii) 1 (iv) 2

(f) In a series RLC circuit if C is increased resonant frequency
(i) Increases (ii) Decreases (iii) It remains the same (iv) It is zero

(g) What is the type of filter shown in the figure?



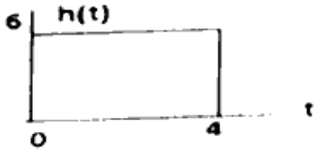
(i) Low pass filter (ii) High pass filter (iii) Band pass filter (iv) All pass filter

(h) Comment on the periodicity of a constant signal?
(i) It is periodic
(ii) It is not periodic
(iii) It is a mixture of period and aperiodic signal
(iv) It depends on the signal

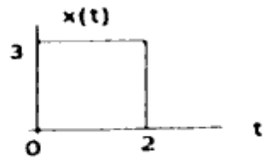
(i) The branches which are not present in a tree is called
(i) twigs, (ii) links, (iii) co-tree (iv) tie-set

(j) A system is linear if it satisfies
(i) Principle of superposition (ii) Principle of homogeneity
(iii) Both (i) & (ii) (iv) Only (i)

Q.2 A given LTI system has produced an impulse response as shown by h(t) given in figure below. Find the time response of this LTI system when an input signal x(t) as shown below is applied to this system. **[14]**



Impulse Response of LTI System



Input to the LTI System

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.3 (a) What is filter? Discuss the advantages of an active filter over a passive filter. [7]

(b) What is band-pass filter? Derive the gain $\left|\frac{V_o}{V_{in}}\right|$ of a first order Butterworth active band-pass filter. [7]

Q.4 (a) Explain clearly with the help of suitable examples the following terms used in network analysis. [10]

a) Oriented graph b) Tree c) Cut Set
d) Tie Set e) Incidence matrix and its properties.

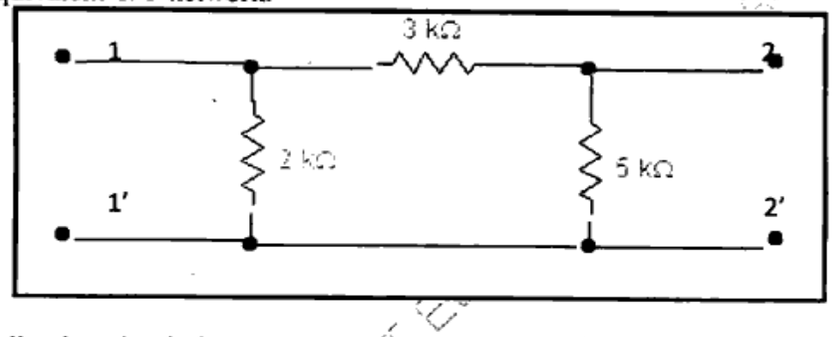
(b) For a symmetrical two port network prove that AD - BC = 1 [4]

Q.5 (a) Obtain the first form of the Foster Network for the driving point impedance of LC network given as: [10]

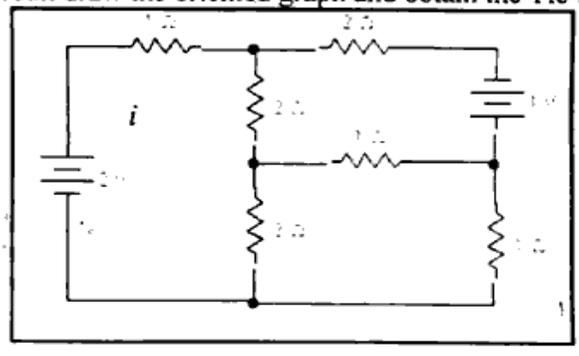
$$Z(S) = \frac{10(S^2+4)(S^2+16)}{S(S^2+9)}$$

(b) What is Cauer's form of circuit realization? Describe briefly. [4]

Q.6 Establish the π - T/ Y - Δ transformation relations. Convert the π/Δ network given below into an equivalent T/Y network. [14]



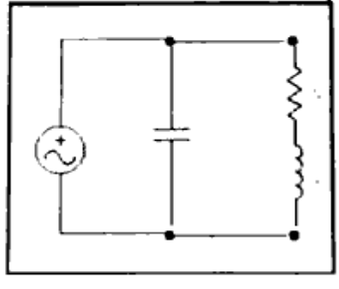
Q.7 For the following circuit draw the oriented graph and obtain the Tie-set matrix. [14]



Q.8 In the following circuit $V_s = 200\sqrt{2}\cos 500t$, R=80 Ω, L=160 mH. [14]

(a) Determine the current drawn from V_s and power absorbed by the LR circuit.

(b) A capacitor is then connected across the V_s so that the current now drawn from V_s is in phase with V_s. Determine the value of the capacitor and the power drawn from V_s.



Q.9 Write short notes on *any two* of the following: [7x2=14]

(a) Active Band pass filter (b) causal & non causal signals with examples
(c) Zero order hold circuit (d) Aliasing and its effect

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play